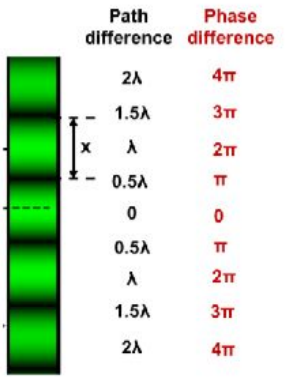
	ratio = 2.6/4.0 = 0.65	A1
5a	When two (or more) waves meet at a point, the resultant displacement is equal to the vector sum of the individual displacements.	M1 A1
5bi	The sources are in-phase at the double-slit (Since path difference (from slits to A) is zero) the phase difference is zero / the waves meet in-phase so constructive interference at A.	B1 B1
5bii	$x = \frac{\lambda D}{a}$ $= \frac{(546 \times 10^{-9})(0.85)}{0.13 \times 10^{-3}} = 3.57 \times 10^{-3}\,\text{m}$ AB = 2.5x = 2.5 (3.57 x 10^{-3})	C1 C1

	= 8.9 x 10^{-3} m  	A1
5biii	The wavelength for blue light is shorter, so the separation is less.	B1
5biv	Maxima/ bright fringe(s)/ central bright fringe/ fringe at A is less bright Minima/ dark fringe(s) is brighter Contrast between maxima and minima (or fringes) decreases	B2
6a	field strength (at a point) is equal to potential gradient / gradient of potential-distance	B1